

**9 (long)** A 4-bit synchronous down counter starts from 15 (1111 in binary) and counts downwards to 0 (0000 in binary) and starts a new cycle again. It is to be designed using J-K bistables.

- (a) Write down the complete truth table. [6]
- (b) Construct the Karnaugh maps and derive the logic expressions for the J and K inputs. [6]
- (c) Draw the complete circuit diagram. [12]
- (d) Draw the signal at the bistable outputs during 10 clock cycles for a 1 GHz clock frequency. [6]